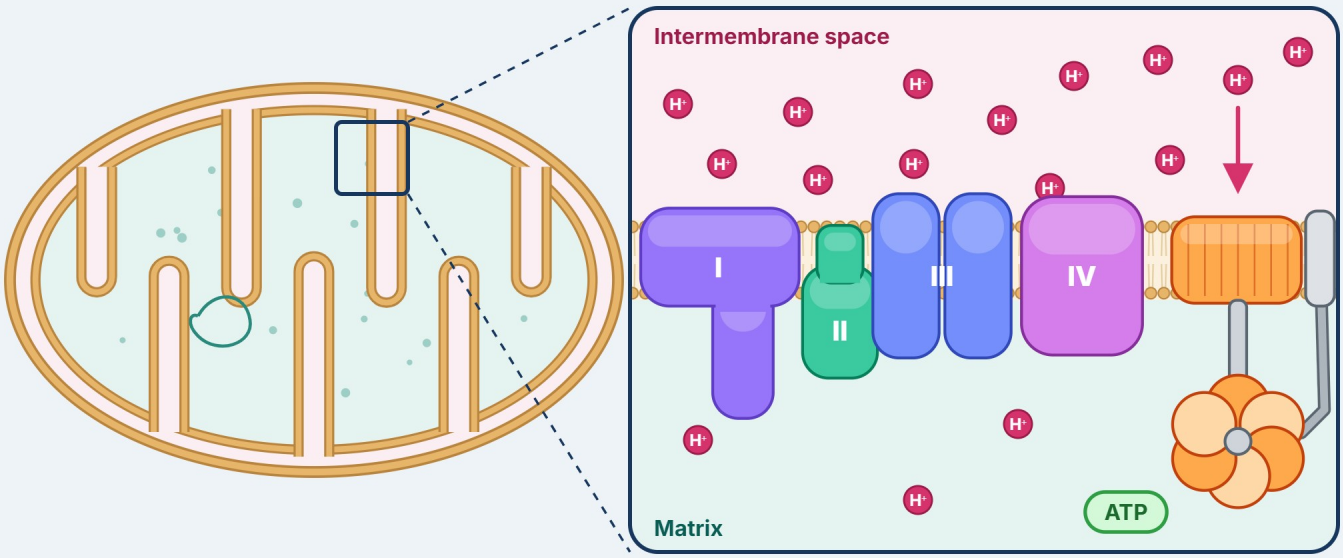
The Electron Transport Chain

and oxidative phosphorylation



Each mitochondrion is folded into thousands of cristae. Zoom into the inner membrane and you find the machinery that makes most of your ATP: four protein complexes (I–IV) and ATP synthase.

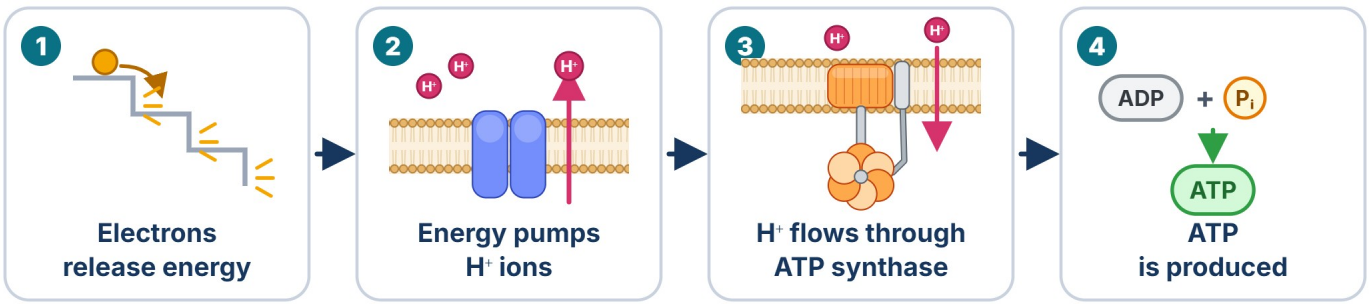
IN THIS CHAPTER YOU WILL LEARN TO

1. explain how NADH and $FADH_2$ link the Krebs cycle to the electron transport chain;
2. locate the electron transport chain in the inner mitochondrial membrane;
3. explain how energy released by moving electrons is used to pump hydrogen ions and build a proton gradient;
4. describe how ATP synthase uses the proton gradient to make ATP (chemiosmosis);
5. explain why oxygen is essential as the final electron acceptor;
6. account for the approximate ATP yield of aerobic respiration.

KEY TERMS

electron transport chain
electron carrier
proton gradient
ATP synthase
chemiosmosis
oxidative phosphorylation
final electron acceptor

THE BIG IDEA OF THIS CHAPTER



Builds on Chapter 6: the Krebs cycle **Leads to** fermentation and anaerobic respiration

7.1 From the Krebs cycle to the electron transport chain

In Chapter 6 you followed glucose through glycolysis, the link reaction and the Krebs cycle. Surprisingly little ATP has been made so far. Most of the energy that was in glucose is now carried by two small molecules: **NADH** and **FADH2**.

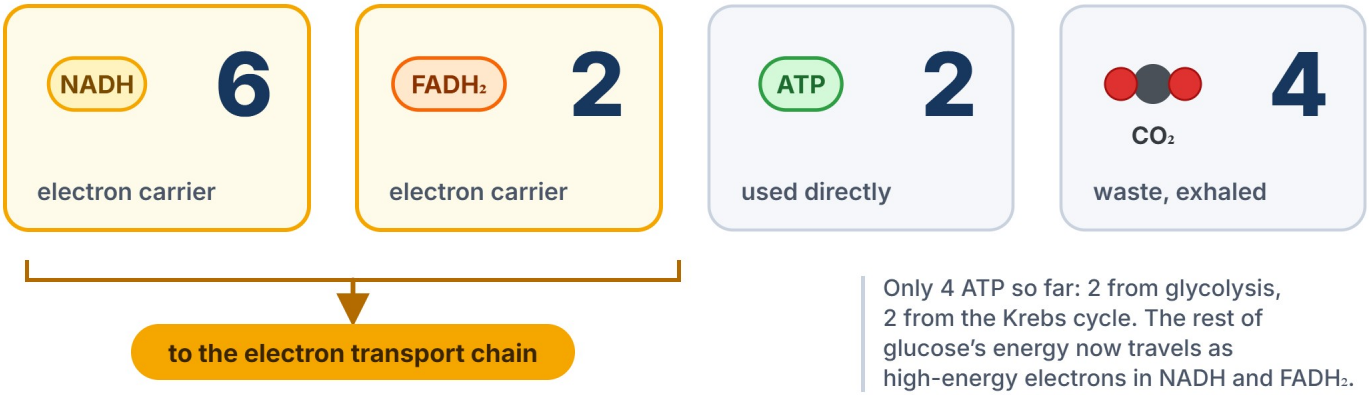


Figure 7.1 Products of the Krebs cycle for one molecule of glucose (two turns of the cycle).

NADH and FADH2 are **electron carriers**. During the earlier stages of respiration they collect high-energy electrons removed from glucose and transport them to the next stage, the **electron transport chain** (ETC).

A carrier that gains electrons is *reduced* ($NAD^+ \rightarrow NADH$). When it hands its electrons to the chain it is *oxidised* back to NAD^+, ready to collect more. In this way a small pool of carriers can shuttle electrons again and again (Figure 7.2).

The electron transport chain releases the energy of these electrons in small, controlled steps and uses it to make most of the cell's ATP: about 26–28 of the 30–32 ATP produced from each molecule of glucose.

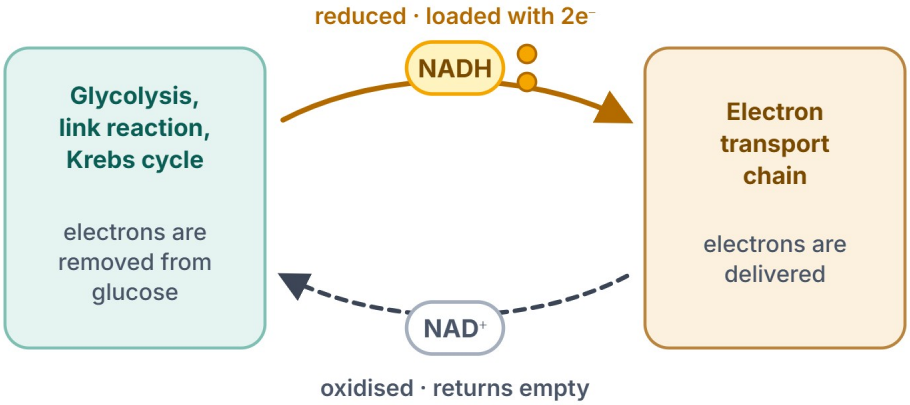


Figure 7.2 Electron carriers work like rechargeable shuttles. FAD and FADH2 cycle in the same way.

KEY TERMS

Electron carrier
A molecule that accepts electrons in one reaction and donates them in another (NAD^+, FAD).

Oxidation
Loss of electrons.

Reduction
Gain of electrons.

MEMORY AID

OIL RIG

Oxidation **I**s **L**oss, **R**eduction **I**s **G**ain of electrons.

CHECK YOUR UNDERSTANDING

1. What do NADH and FADH2 do?
2. Is NADH the reduced or the oxidised form?

7.2 Where it happens: the inner membrane

A mitochondrion has **two membranes** (Figure 7.3). The smooth outer membrane surrounds the organelle. The inner membrane is folded into many **cristae**, which greatly increase its surface area. Between the two lies the narrow **intermembrane space**; inside the inner membrane is the **matrix**, where the Krebs cycle takes place.

The electron transport chain is located in the **inner mitochondrial membrane**. Its proteins transfer electrons and pump hydrogen ions. The membrane is also almost impermeable to H^+, which turns out to be essential (section 7.5).

KEY TERMS

Cristae
Folds of the inner membrane; more folds, more ETC proteins.

Matrix
Fluid inside the inner membrane.

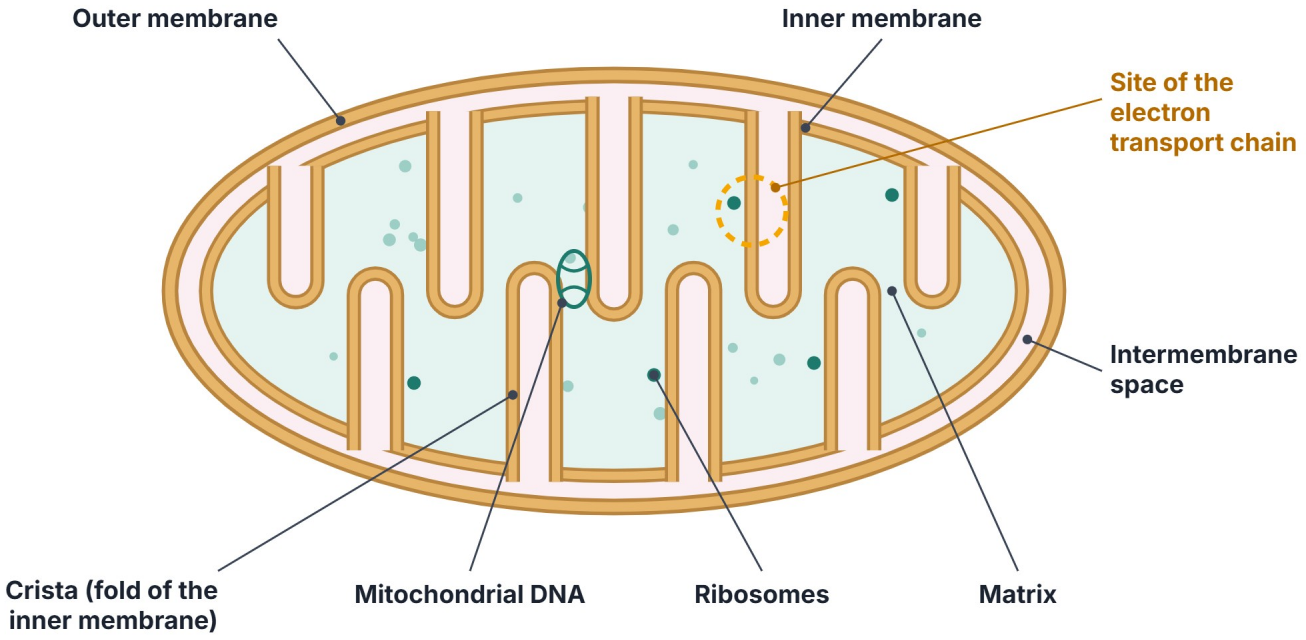


Figure 7.3 A mitochondrion cut lengthwise. The electron transport chain is built into the inner membrane, so cells that need a lot of ATP (muscle, heart) have mitochondria with densely packed cristae.

HOW TO READ THE DIAGRAMS IN THIS CHAPTER

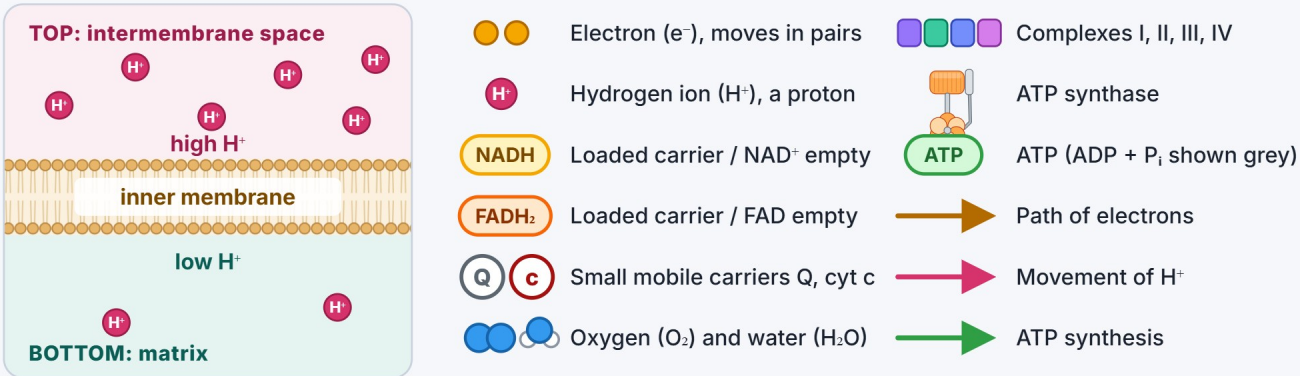


Figure 7.4 All membrane diagrams show the intermembrane space at the top and the matrix below. Colours never change meaning.

7.3 Step 1 – Electrons enter the transport chain

NADH and FADH2 deliver their high-energy electrons to the chain. From there, electrons are handed from one protein to the next, like a bucket passed along a line of people.

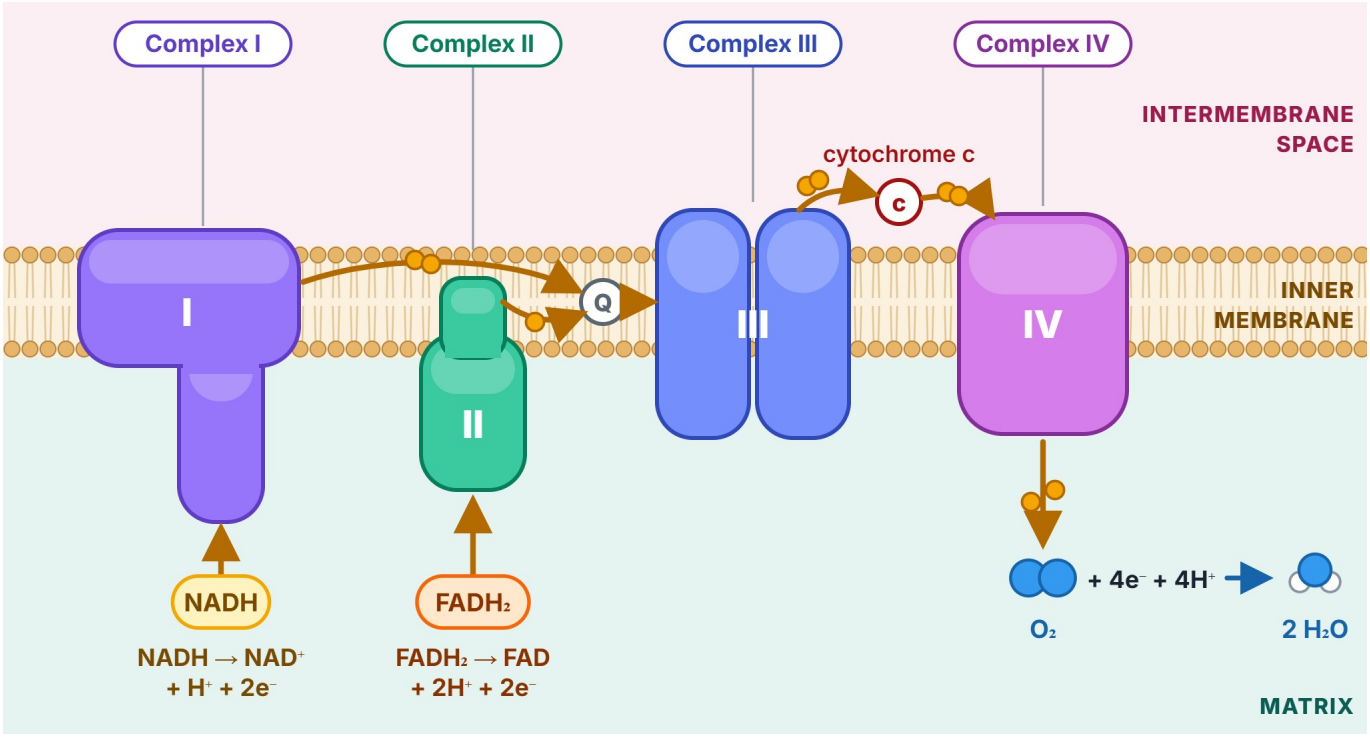


Figure 7.5 Electrons enter the chain. NADH donates electrons to Complex I; $FADH_2$ donates them to Complex II. Gold arrows trace the electron path. At Complex IV electrons are passed to oxygen, which forms water.

NADH gives its pair of electrons to the first protein, **Complex I**. In doing so NADH is oxidised to NAD^+, which returns to the matrix to be reloaded.

$FADH_2$ gives its electrons to **Complex II**, so it **bypasses Complex I**. ($FADH_2$ is actually formed inside Complex II, which is also an enzyme of the Krebs cycle.)

Electrons then move to Complex III and finally Complex IV, ferried between the complexes by two small mobile carriers, ubiquinone (Q) and cytochrome c.

	NADH	$FADH_2$
Enters at	Complex I	Complex II
Route	I → III → IV	II → III → IV
Oxidised to	NAD^+	FAD
H^+ pumped per $2e^-$	10	6
ATP made (approx.)	≈ 2.5	≈ 1.5

COMMON MISCONCEPTION

"Electrons are used up in the chain."

No. Electrons are **passed along**. Only their energy is used. At the end they are accepted by oxygen.

DO I NEED TO KNOW Q AND CYTOCHROME C?

Not by name. Focus on the route: **I → III → IV** for NADH and **II → III → IV** for $FADH_2$.

7.4 Electrons release energy step by step

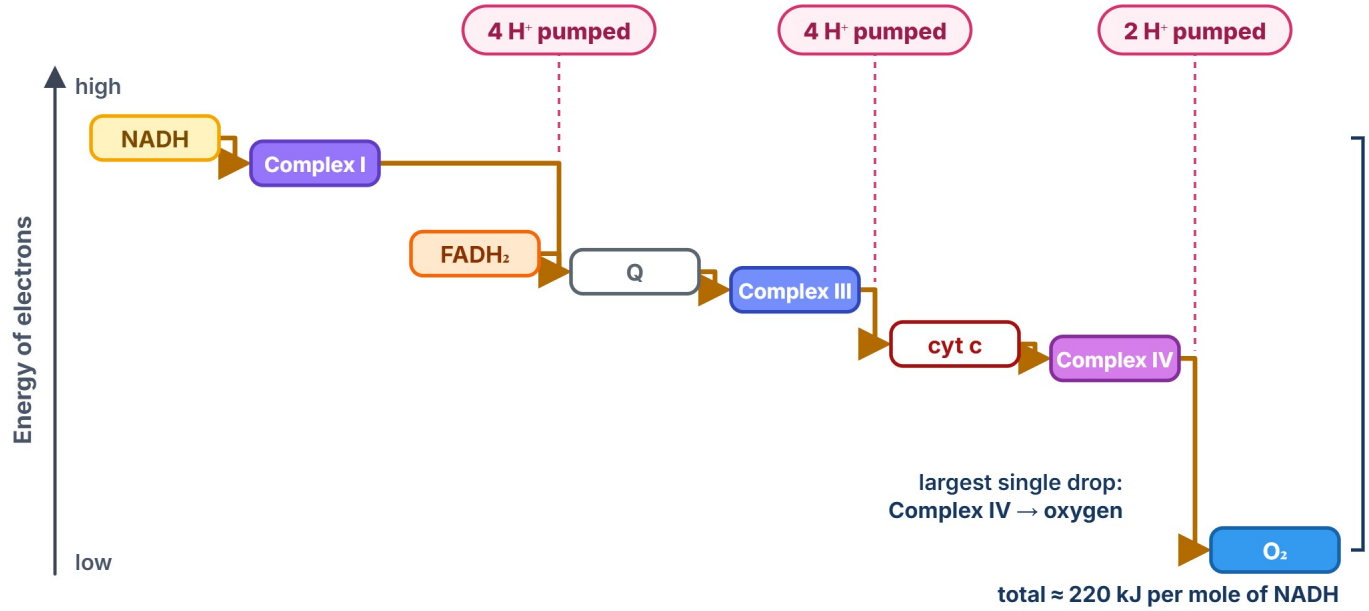


Figure 7.6 The electron energy staircase. Levels are based on how strongly each carrier holds electrons. Energy released at three steps (pink labels) is used to pump hydrogen ions. Numbers are H^+ pumped per pair of electrons.

As electrons pass from one carrier to the next, each new carrier attracts electrons a little more strongly than the one before. The electrons therefore move **downhill in energy**, and each step releases a small amount of energy (Figure 7.6). Oxygen, at the bottom of the staircase, attracts electrons most strongly of all.

Complexes I, III and IV use the energy released at their steps to pump hydrogen ions across the inner membrane. Complex II sits on a much smaller step and pumps none, which is why $FADH_2$ yields less ATP than NADH.

If NADH reacted with oxygen directly, the energy would be released in one burst, mostly as heat. A chain of carriers lets the cell capture it in usable portions (Figure 7.7).

WHY A CHAIN?

Small steps let each complex capture a usable amount of energy. A single big step would waste most of it as heat.

CHECK YOUR UNDERSTANDING

1. Which step in Figure 7.6 releases the most energy?
2. Why does $FADH_2$ lead to fewer H^+ being pumped than NADH?

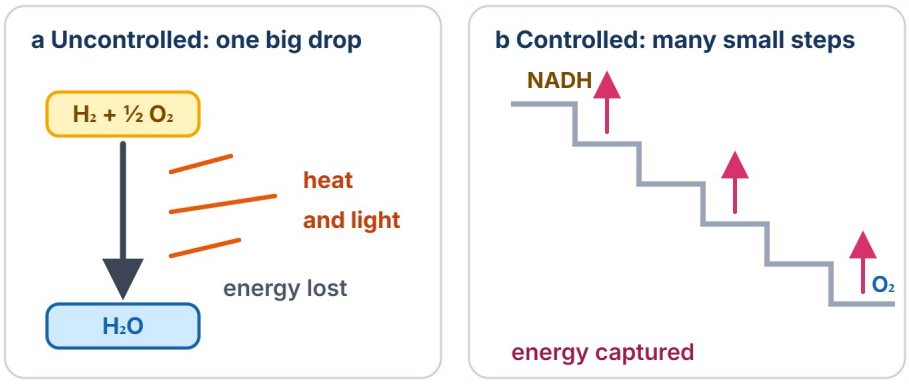


Figure 7.7 Burning hydrogen releases its energy at once (a); the ETC releases it in usable steps (b).

7.5 Step 2 – Pumping hydrogen ions builds a gradient

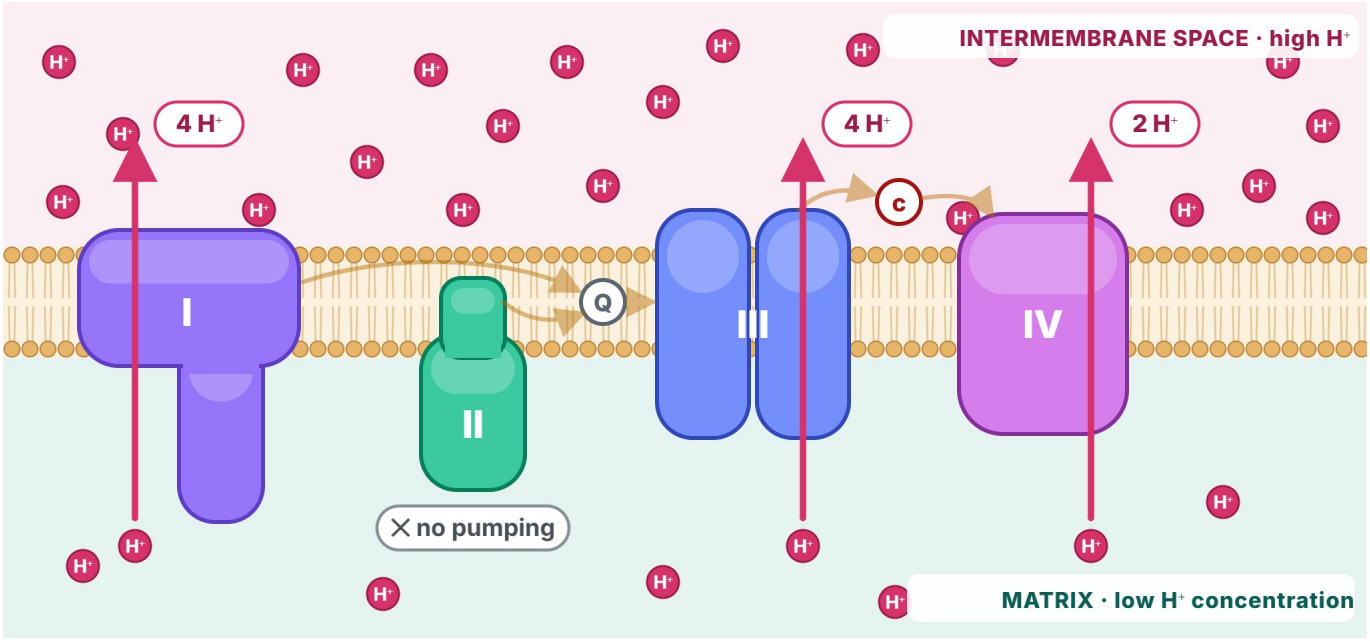


Figure 7.8 Energy from moving electrons (faint gold path) drives H^+ pumping (pink arrows) at Complexes I, III and IV. H^+ accumulates in the intermembrane space.

Each time electrons pass through Complexes I, III and IV, the energy released is used to pump **hydrogen ions (H^+)** from the matrix into the intermembrane space. Complex II does not pump H^+.

As a result, many hydrogen ions accumulate in the intermembrane space while fewer remain in the matrix. This difference in H^+ concentration across the inner membrane is called the **proton gradient**. Like water held behind a dam, the gradient stores **potential energy**.

The H^+ ions tend to move back into the matrix, but they cannot cross the inner membrane freely. There is only one way back: through ATP synthase (section 7.6).

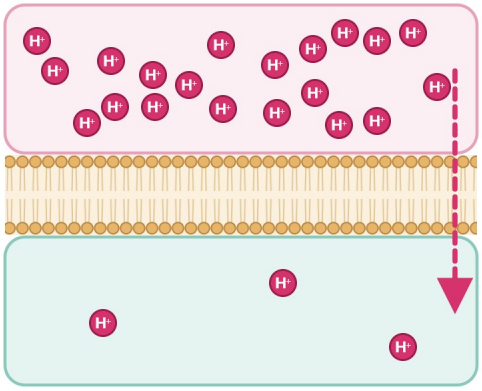


Figure 7.9 The proton gradient. The inner membrane separates a region of high H^+ concentration from one of low concentration.

WHICH COMPLEXES PUMP?

Complex	Pumps?	H^+
I	✓ yes	4
II	✕ no	0
III	✓ yes	4
IV	✓ yes	2

KEY TERM

Proton gradient
A difference in H^+ concentration (and charge) across a membrane that stores potential energy.

H^+ = proton.

QUICK CHECK

Where is the H^+ concentration higher: matrix or intermembrane space?

7.6 Step 3 – ATP synthase makes ATP

The hydrogen ions in the intermembrane space tend to flow back into the matrix, where their concentration is lower. They cannot cross the lipid bilayer, so they pass through a special protein: **ATP synthase**.

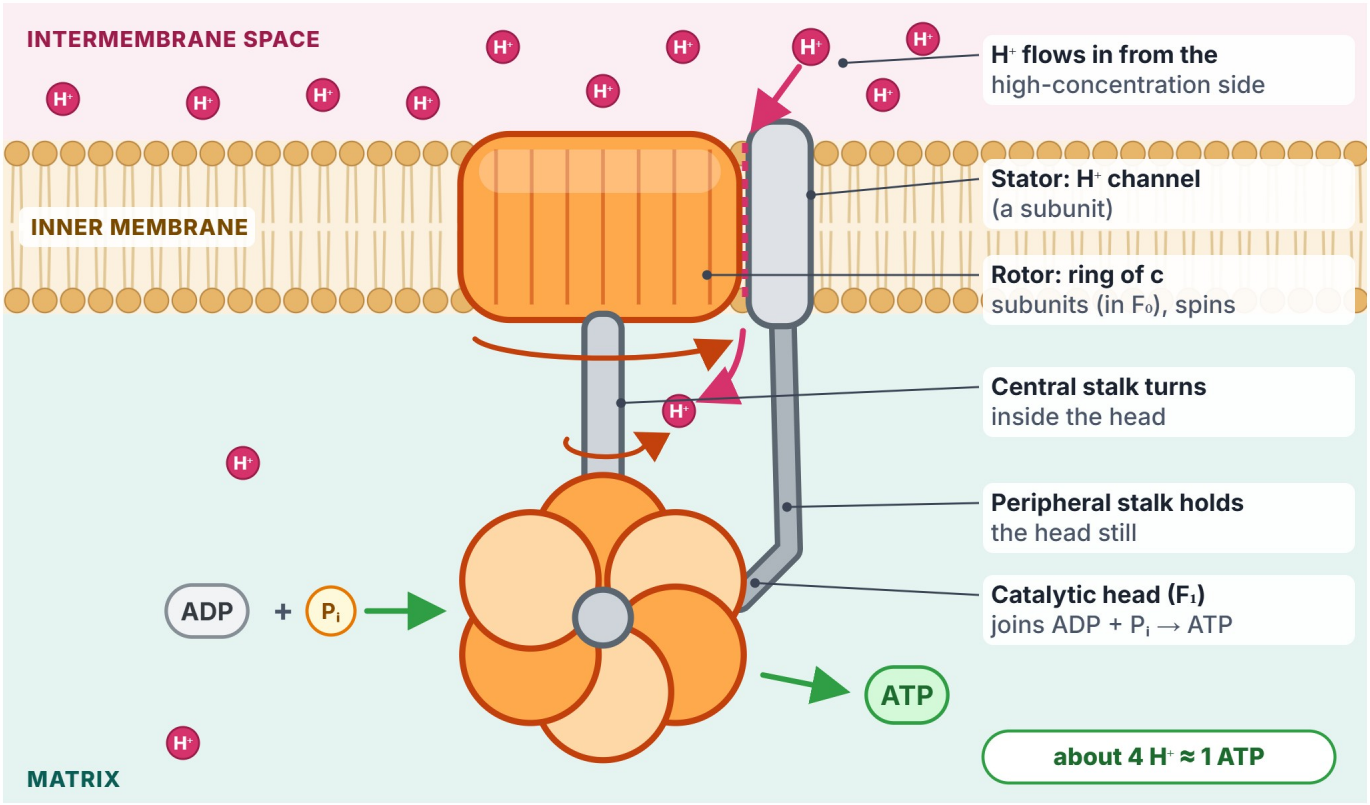


Figure 7.10 Structure of ATP synthase. H^+ moves down its concentration gradient through the channel (pink), turning the rotor and stalk (orange arrows). The head converts ADP + P_i into ATP (green).

ATP synthase works like a tiny rotary motor. As H^+ ions flow through its channel, they turn the **rotor** in the membrane. The rotor turns the **central stalk**, which changes the shape of the **catalytic head**. Each change in shape joins ADP and inorganic phosphate (P_i) to make ATP (Figure 7.11).

Roughly four H^+ ions flow back for every ATP made, including those used to bring phosphate into the matrix.

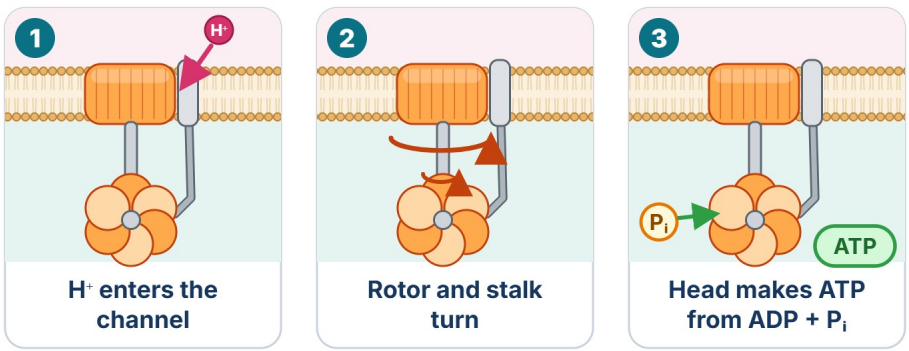


Figure 7.11 ATP synthase in three steps. The head does not rotate; the stalk turns inside it.

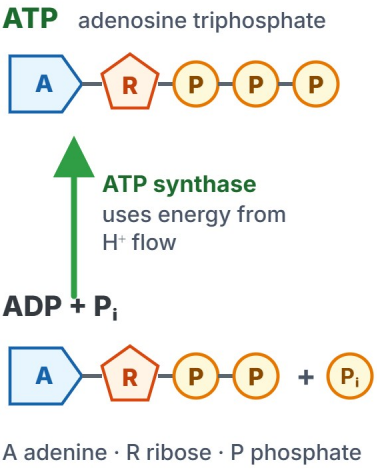


Figure 7.12 ATP has one more phosphate group than ADP. Joining them is a condensation reaction (water is released).

7.7 Chemiosmosis: the power-station analogy

A hydroelectric power station turns stored water into electricity. A mitochondrion turns a stored H^+ gradient into ATP. The two work in the same way.

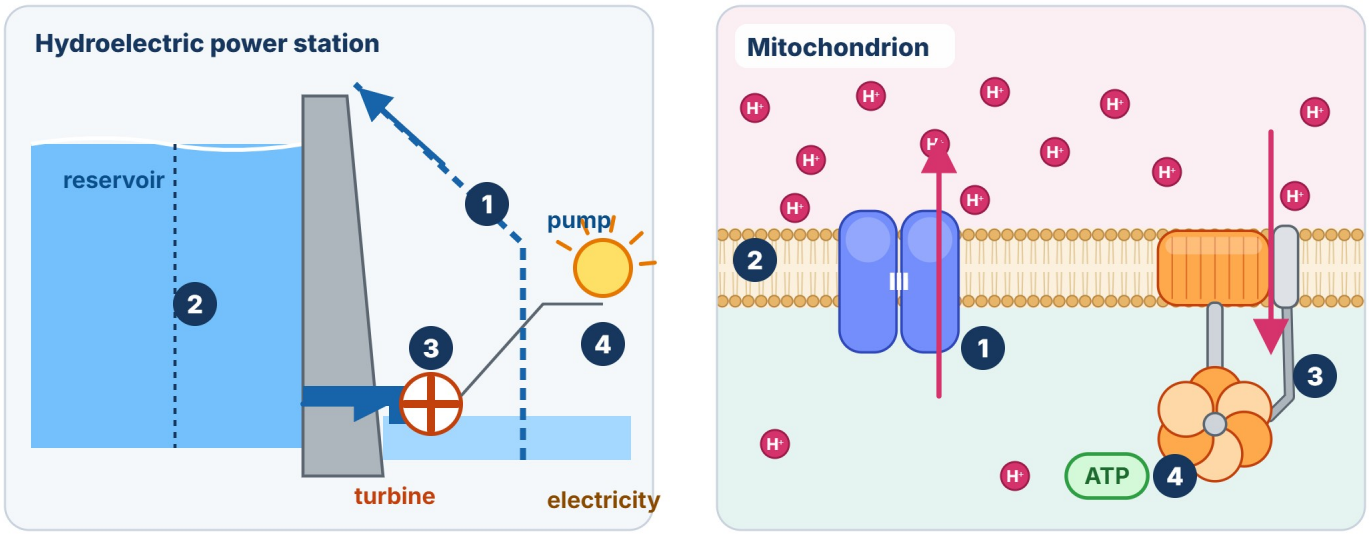


Figure 7.13 Matching numbers show matching parts. Electron transport "pumps up the reservoir"; ATP synthase is the turbine.

#	Hydroelectric station	Mitochondrion
1	Water is pumped up into the reservoir	Complexes I, III, IV pump H^+ into the intermembrane space
2	Difference in water level stores energy	The **proton gradient** stores potential energy
3	Water flows down through the turbine	H^+ flows back through **ATP synthase**
4	The turning turbine produces electricity	ATP synthase turns and produces **ATP**

CHEMIOSMOSIS

Making ATP using the energy of an H^+ gradient: H^+ flows through ATP synthase and drives ADP + P_i → ATP.

OXIDATIVE PHOSPHORYLATION

The whole process: electron transfer in the ETC builds the gradient, and chemiosmosis uses it to make ATP. It needs oxygen.

REMEMBER

"The proton gradient provides the energy that drives ATP synthase to produce ATP."

SCIENCE HISTORY

Peter Mitchell proposed chemiosmosis in 1961. Many biochemists doubted it, but the evidence won: he received the Nobel Prize in Chemistry in 1978.

WHERE THE ANALOGY BREAKS DOWN

A dam turns energy into electricity; a mitochondrion stores it in ATP. And the "pumps" are powered by electrons, not by electricity.

7.8 Step 4 – Why is oxygen necessary?

Electrons have passed all the way through the chain. But where do they go at the end?

OXYGEN ACCEPTS ELECTRONS AT COMPLEX IV

$$O_2 + 4e^- + 4H^+ \longrightarrow 2\,H_2O$$

O_2	$4e^-$	$4H^+$	$2\,H_2O$
final electron acceptor	from Complex IV	from the matrix	water (a product)

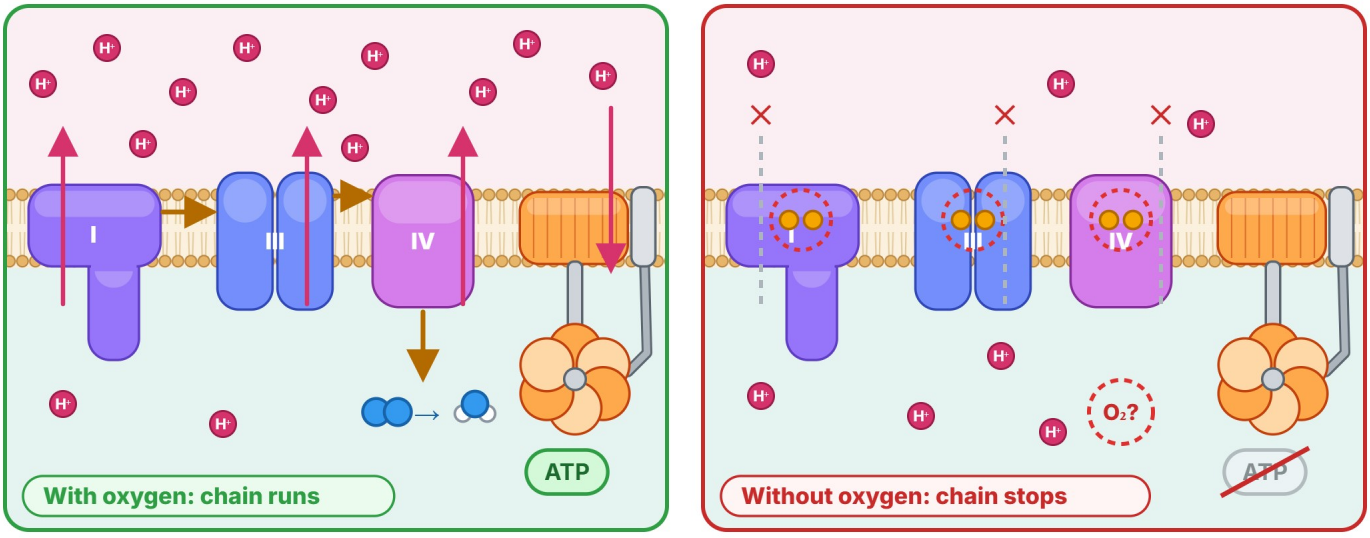


Figure 7.14 The chain with oxygen (left) and without it (right). Without a final acceptor, electrons pile up at the complexes, H^+ pumping stops, the gradient runs down and ATP synthase falls idle.

FIGURE 7.15 · WHAT HAPPENS WITHOUT OXYGEN

1. Oxygen cannot accept electrons
2. Electrons can no longer move through the chain
3. Proton pumping stops
4. The proton gradient cannot be maintained
5. ATP production by oxidative phosphorylation stops

At the end of the chain, Complex IV passes electrons to **oxygen**. Oxygen is the **final electron acceptor**: it combines with the electrons and with hydrogen ions from the matrix to form water.

Without oxygen the electrons have nowhere to go, so the whole chain backs up like traffic behind a closed exit (Figure 7.15). Cells must then rely on glycolysis and fermentation, which give only 2 ATP per glucose.

REMEMBER

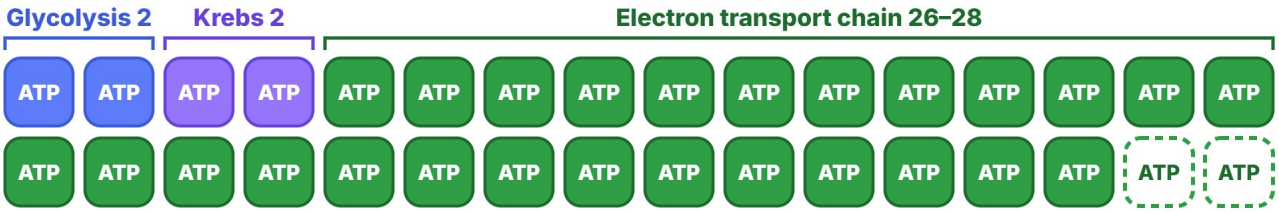
Oxygen does **not** make ATP directly. It keeps the chain moving by removing electrons at the end.

BIOLOGY IN CONTEXT

Cyanide blocks Complex IV. Electrons cannot reach oxygen, ATP production stops within minutes, and the poison is deadly.

7.9 How much ATP does one glucose make?

The electron transport chain produces most of the ATP in aerobic respiration: roughly 9 out of every 10 molecules.



Dashed tokens: the 2 ATP that some cells do not get (see "Why a range?").

Figure 7.16 Every square is one ATP. The ETC (green) makes about 87% of the total.

Source	ATP per glucose
Glycolysis	2
Krebs cycle	2
Electron transport chain	26–28
Total	**30–32**

TOTAL FROM ONE GLUCOSE

30–32 ATP

a typical estimate for eukaryotic cells

NADH and $FADH_2$ do not make ATP directly. They carry electrons to the ETC; the energy of those electrons builds the proton gradient, and the gradient drives ATP synthase. So we can estimate the ATP yield from the number of carriers.

Worked example: counting the ATP

Electron carrier	per glucose	≈ ATP each	ATP
NADH	10	2.5	25
$FADH_2$	2	1.5	3
ETC total			**28**
+ glycolysis + Krebs cycle			4
Total per glucose			**32**

Each NADH leads to 10 H^+ being pumped and each $FADH_2$ to 6. With about 4 H^+ needed per ATP, that gives ≈ 2.5 and ≈ 1.5 ATP. These are averages, not whole-number reactions, so textbook totals are given as a range.

CHECK YOUR UNDERSTANDING

Why do NADH and $FADH_2$ not count as "making ATP" themselves?

WHERE THE 10 NADH COME FROM

Glycolysis	2
Link reaction	2
Krebs cycle	6

The Krebs cycle also gives the 2 $FADH_2$.

WHY A RANGE?

The 2 NADH from glycolysis are made in the cytoplasm. In some cells (e.g. muscle, brain) their electrons enter the mitochondrion via FAD, so they are worth ≈ 1.5 ATP each. Then the ETC gives 26 and the total is 30.

7.10 Putting it all together

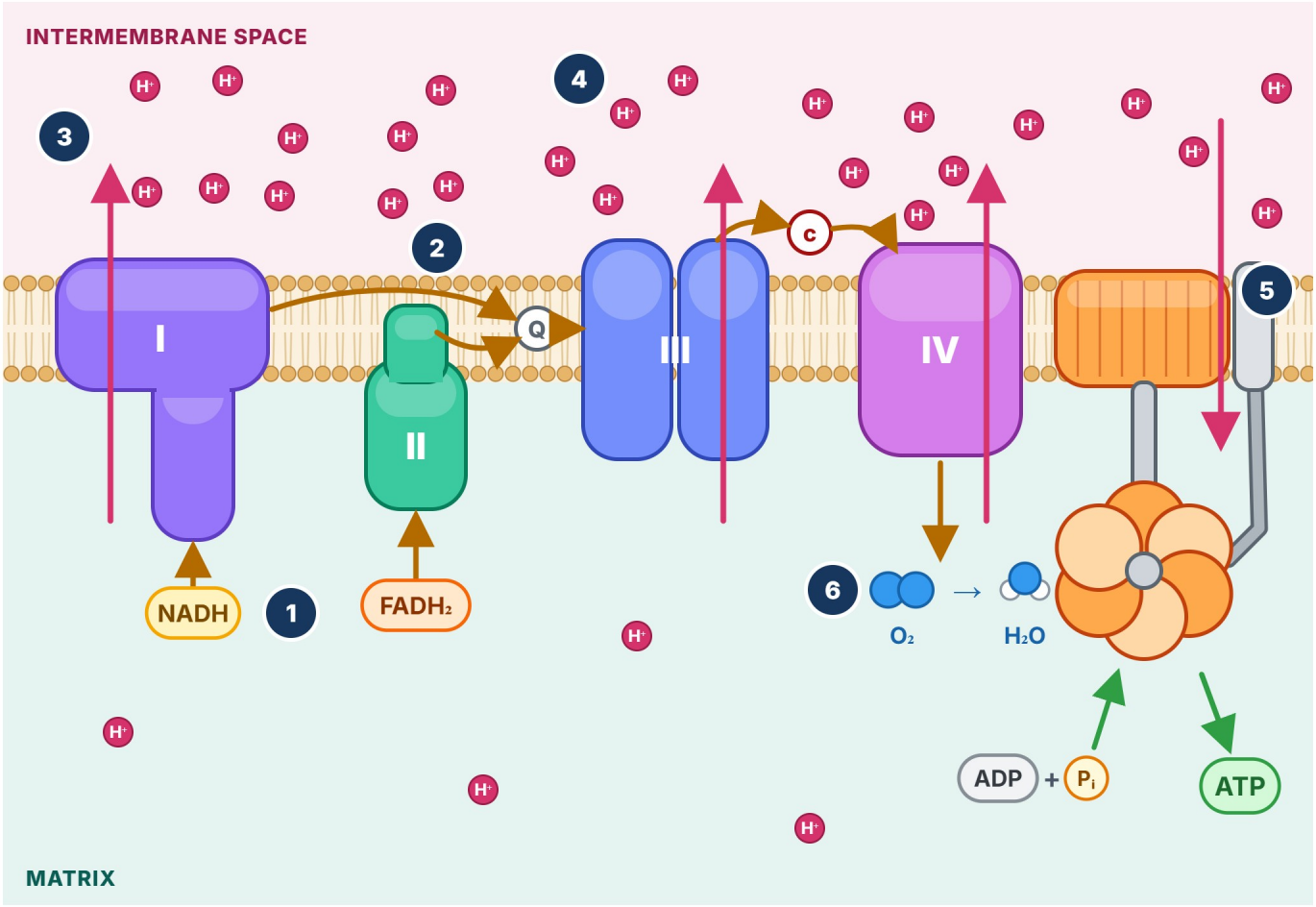


Figure 7.17 The complete picture of oxidative phosphorylation. Copy this diagram into your notebook, building it up in the order of the numbered steps.

1 NADH and $FADH_2$ deliver electrons
Electrons enter Complexes I and II in the inner membrane.

2 Electrons move through the chain
Energy is released as electrons are transferred.

3 Hydrogen ions are pumped
Complexes I, III and IV pump H^+ into the intermembrane space.

4 A proton gradient develops
The difference in H^+ concentration stores potential energy.

5 ATP synthase produces ATP
H^+ flows back into the matrix through ATP synthase.

6 Oxygen accepts electrons
Oxygen combines with electrons and H^+ to form water.

THE ONE SENTENCE TO REMEMBER

Electrons provide the energy to pump hydrogen ions, and the movement of hydrogen ions through ATP synthase provides the energy to produce ATP.

7.11 Chapter review

A · MULTIPLE CHOICE

1. Where does the electron transport chain occur?

a) Mitochondrial matrix
b) Inner mitochondrial membrane
c) Cytoplasm

2. What is the function of NADH and $FADH_2$?

a) Carry electrons
b) Produce glucose
c) Break down ATP

3. Which complexes pump hydrogen ions?

a) I, II and III
b) II, III and IV
c) I, III and IV

4. What is ATP synthase responsible for?

a) Producing ATP
b) Producing CO_2
c) Breaking down glucose

5. What is the final electron acceptor?

a) Carbon dioxide
b) Oxygen
c) Glucose

6. What happens when oxygen is unavailable?

a) The ETC stops functioning
b) ATP production increases
c) More oxygen is produced

B · LABEL THE DIAGRAM

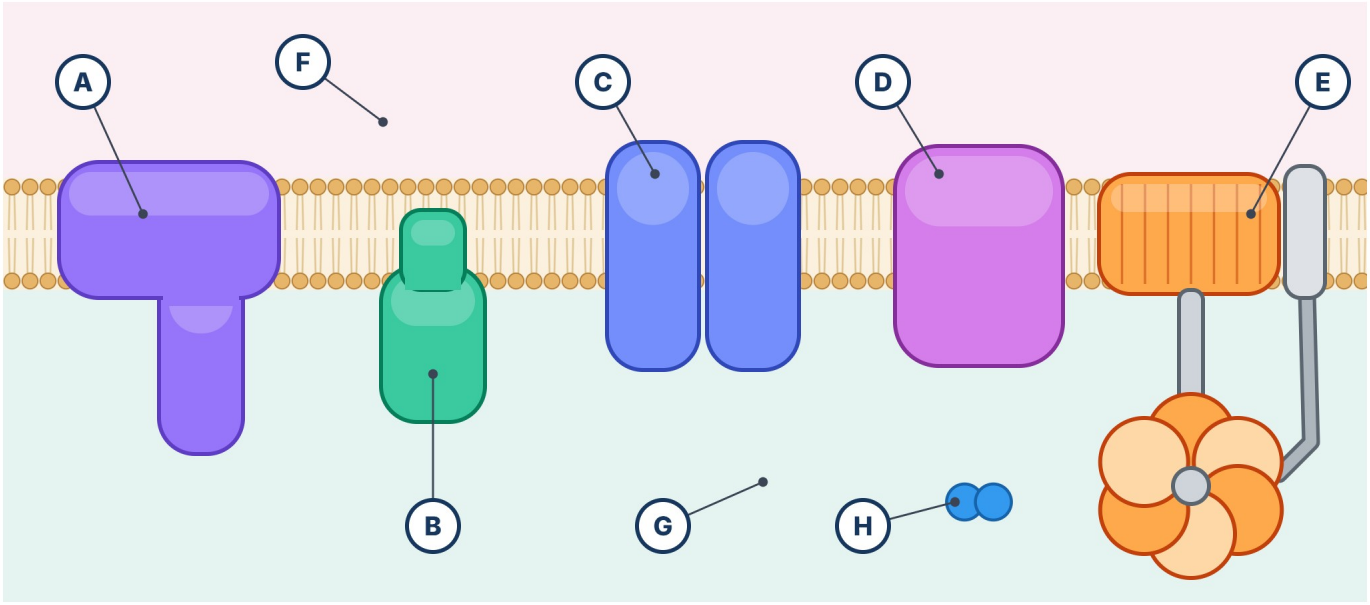


Figure 7.18 Write the correct term for each letter A–H.

Complex I | Complex II | Complex III | Complex IV | ATP synthase | intermembrane space | matrix | oxygen

C · SHORT ANSWER

1. Explain why $FADH_2$ leads to less ATP than NADH. [2]
2. Use the hydroelectric dam analogy to explain the role of the proton gradient. [3]
3. A poison blocks Complex IV. Predict and explain its effect on (a) H^+ pumping and (b) ATP production. [4]
4. Explain the statement "Oxygen does not directly produce ATP." [2]

D · EXTENSION

Brown fat cells contain a protein that lets H^+ leak back into the matrix **without** passing through ATP synthase. Predict what happens to (a) ATP production and (b) heat production in these cells, and suggest why newborn babies have brown fat. [4]

Answers A: 1 b · 2 a · 3 c · 4 a · 5 b · 6 a **B:** A Complex I · B Complex II · C Complex III · D Complex IV · E ATP synthase · F intermembrane space · G matrix · H oxygen